PROJECT REPORT

Incident Lifecycle Automation in ServiceNow

A Project Report

Submitted by

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1. Introduction

1.1 Project Overview

Incident Lifecycle Automation in ServiceNow is a project that defines and validates a structured Incident Management lifecycle on the ServiceNow platform. The lifecycle covers incident creation, classification, knowledge-assisted investigation, assignment and escalation, Level 2 support, emergency change linkage, child incidents, resolution and closure, knowledge creation, and final validation.

In IT Service Management, an incident is an unplanned interruption to an IT service or a reduction in its quality. ServiceNow Incident Management gives a service desk a single record on which the issue is logged, classified, assigned, investigated, resolved, and closed. By linking the incident with the affected Service, Service Offering, and Configuration Item (CI), and with related Change, Child Incident, Knowledge, and SLA records, ServiceNow keeps the context needed to handle the issue from creation to closure.

This report documents the designed lifecycle, its roles, functional requirements, architecture, validation approach, and configuration reference. As stated in the original project documentation, the report describes the **designed/planned solution**. Actual execution evidence and screenshots are to be added in the marked placeholders after implementation and validation (see Section 1.8).

1.2 Purpose

The purpose of the project is to provide a consistent reference for the planned solution, its functional flow, roles, related records, validation activities, and deployment-readiness considerations. The solution is needed for the following reasons:

- **Standardized incident handling** – a defined lifecycle makes every incident follow the same sequence of creation, classification, investigation, resolution, and closure, whichever agent handles it.
- **Better incident classification** – Category, Subcategory, Urgency, Service, and Configuration Item are captured consistently so that incidents can be routed and analysed reliably.
- **Knowledge-assisted troubleshooting** – Knowledge Management and Agent Assist give agents access to existing solutions during investigation instead of diagnosing known issues from the beginning.
- **Improved assignment and escalation** – clear assignment and escalation to Level 2 support helps complex incidents reach the right technical team with their context intact.
- **Level 2 investigation** – Level 2 Technicians receive the existing incident context, investigate the technical issue, and record findings on the same incident.
- **Related-record traceability** – Emergency Change Requests, Child Incidents, knowledge, and SLA information stay traceable from the incident.
- **Complete resolution documentation** – Cause, Resolution, Resolution Details, and closure information are documented so the record supports review and reuse.

1.3 Problem Background

When incident handling is inconsistent, the quality of service depends on individual habits rather than on a defined process. The following challenges are commonly associated with informal or inconsistent incident management. They are given as general background to the project and are not measurements taken from an implemented system.

- Incomplete information at the time of logging makes triage and routing harder.
- Inconsistent classification of Category, Urgency, Service, and CI leads to misrouting and unreliable reporting.
- Agents repeat investigation of known issues when troubleshooting knowledge is not captured or is difficult to find.

- Escalated incidents lose context, so Level 2 Technicians repeat questions that were already answered.
- Emergency changes and recurring or related issues are not linked to the incident, which weakens traceability.
- Resolution notes are incomplete, so lessons are not reused and closed incidents are hard to review.
- Limited visibility of SLA information and related records makes it harder to prioritise work.

This project responds to these difficulties by describing a single structured lifecycle in ServiceNow in which each stage has defined inputs, responsible roles, and related records.

1.4 Project Objectives

1. Implement a structured Incident Management lifecycle from creation to closure in ServiceNow.
2. Set up Service and Service Offering records and associate them with incidents.
3. Capture complete incident information and classify each incident consistently using Category, Subcategory, Urgency, Service, and Configuration Item.
4. Support troubleshooting with Knowledge Management and Agent Assist.
5. Assign and escalate incidents to Level 2 support while retaining incident context.
6. Link Emergency Change Requests and Child Incidents to the incident where required.
7. Document Cause, Resolution, Resolution Details, and closure information, and convert useful troubleshooting information into Knowledge Articles.
8. Verify SLA and related-record visibility, and validate the complete lifecycle through testing and deployment validation.

1.5 Project Scope

The scope of the project, as defined in the original documentation, is:

- Incident Management lifecycle from creation through closure.
- Service and Service Offering setup and association.
- Incident creation and classification using Category, Subcategory, Urgency, Service, and Configuration Item (CI).
- Knowledge Management and Agent Assist support.
- Assignment and escalation to Level 2 support.
- Emergency Change Request linkage.
- Parent and Child Incident relationships.
- SLA and related-record visibility.
- Resolution, closure information, and Knowledge Article creation.
- Final validation of incident, SLA, change, child incident, knowledge, and related records.
- Testing and deployment validation.

Documented boundaries: the source documentation does not describe an external dataset, custom source code, external system integration, or a machine-learning component. These are therefore outside the documented scope of this report.

1.6 Users and Roles

Five roles take part in the lifecycle. Their responsibilities, taken from the original documentation and the User Guide, are summarised in Table 1.1.

Table 1.1: Users, roles, and responsibilities

Role	Responsibilities
Service Desk Agent	Creates, classifies, assigns, investigates, and updates incidents. Uses Knowledge and Agent Assist during troubleshooting, links an Emergency Change or creates/links a Child Incident when appropriate, records Cause, Resolution, Resolution Details, and closure information, and validates the incident and related records before closure.
Level 2 Technician	Investigates escalated incidents and records technical findings. Reviews the existing context, related knowledge, and associated records; updates findings and resolution information; coordinates related change or child-incident activity when required; and returns the incident for resolution/closure.
Service Owner / Support Stakeholders	Provide service context and support oversight for the Service and Service Offering associated with the incident.
Change Stakeholders	Review and manage emergency changes when required: review emergency change requests linked to incidents, validate the change context and its relationship to the incident, and support change completion and traceability.
Project Team	Validates the lifecycle and deployment readiness, including testing, verification of related records, and capture of user acceptance evidence.

1.7 Expected Benefits

The benefits in Table 1.2 are the expected outcomes of the designed solution. They have not been measured, and no performance figures are claimed.

Table 1.2: Expected benefits of the solution

Expected benefit	Explanation
Standardized Incident Management process	All incidents follow one defined lifecycle with the same steps and records.
Faster handling and improved routing	Consistent classification and clear assignment allow incidents to reach the right team sooner.
Better incident classification and information quality	Category, Subcategory, Urgency, Service, and CI are captured on every incident.
Improved escalation and Level 2 context	Escalated incidents keep their existing context, so Level 2 Technicians can start investigation without repeating questions.
Traceability between incidents, changes, and child incidents	Emergency Changes and Child Incidents are linked to the incident so relationships can be followed.
Complete resolution documentation and reusable knowledge	Cause, Resolution, Resolution Details, and closure information are recorded, and useful information can become a Knowledge Article.
Improved SLA and related-record visibility	SLA information and related records can be viewed and verified from the incident.

1.8 Documentation Status and Evidence Convention

The original documentation states: “This document describes the designed/planned solution. Actual execution evidence, production results, and screenshots should be added after implementation and validation.” It also states that execution results are not asserted because no actual execution evidence was provided with the project documentation materials.

This report follows the same position. Every capability is therefore treated as **Planned** and **Not Verified** until genuine evidence is attached. The status labels in Table 1.3 are used throughout the report, and text in *[square brackets]* marks information or evidence that must be supplied by the project team.

Table 1.3: Status labels used in this report

Label	Meaning
Implemented	The feature is configured and supported by actual evidence such as a screenshot or test record.
Partially Implemented	Part of the feature is configured and evidenced; the remainder is not.
Planned	The feature is described in the design but implementation evidence has not been supplied.
Not Verified	A test or check has not been executed or no evidence is available.
[Placeholder]	Information, screenshot, or value that must be inserted by the project team. It is not evidence.

The Team ID and report date fields appear on the title page as placeholders because no values were present in the supplied document.

2. Ideation Phase

The Ideation Phase records the problems the project sets out to solve and the ideas that led to the lifecycle design. The Ideation Phase documents (problem statements, empathy map, brainstorming board, and idea prioritization) were not part of the supplied files. The structure and the original headings are therefore retained below, and the original content is to be inserted where marked. Nothing has been invented to fill these sections.

2.1 Problem Statement

Two customer problem statements were defined, one for each primary user of the lifecycle: PS-1 for the Service Desk Agent and PS-2 for the Level 2 Technician. Table 2.1 shows the problem context taken from the documented lifecycle requirements and reserves a field for the original wording.

Table 2.1: Customer problem statements

ID	Customer	Problem context (from lifecycle requirements)	Original problem statement
PS-1	Service Desk Agent	Needs to create, classify, assign, and update incidents consistently, with sufficient information for triage and routing, support from knowledge during troubleshooting, and complete resolution and closure information.	*[Insert original PS-1 wording]*
PS-2	Level 2 Technician	Needs escalated incidents to arrive with their existing context, together with related knowledge and records, so that technical investigation can proceed and findings can be recorded.	*[Insert original PS-2 wording]*

2.2 Empathy Map Canvas

The Empathy Map Canvas describes what the Service Desk Agent and Level 2 Technician think, feel, hear, see, say, do, and experience. Its six components are listed in Table 2.2.

Table 2.2: Empathy Map Canvas components

Component	Original content
Think & Feel	*[Insert from original Empathy Map Canvas]*
Hear	*[Insert from original Empathy Map Canvas]*
See	*[Insert from original Empathy Map Canvas]*
Say & Do	*[Insert from original Empathy Map Canvas]*
Pain	*[Insert from original Empathy Map Canvas]*
Gain	*[Insert from original Empathy Map Canvas]*

[Insert original Empathy Map Canvas image here]

Figure 2.1: Empathy Map Canvas (original canvas to be inserted)

2.3 Brainstorming

Brainstorming produced six idea groups. Each group corresponds to a stage of the incident lifecycle, as shown in Table 2.3.

Table 2.3: Brainstorming idea groups and lifecycle alignment

Idea group	Aligned lifecycle stage	Original ideas
Incident Setup	Service and Service Offering setup; incident creation	*[Insert original ideas]*
Classification & Routing	Incident classification; assignment and escalation	*[Insert original ideas]*
Knowledge Support	Knowledge Management and Agent Assist	*[Insert original ideas]*
Related Records	Emergency Change, Child Incident, SLA and related records	*[Insert original ideas]*
Resolution & Knowledge	Resolution and closure; Knowledge Article creation	*[Insert original ideas]*
Validation	Final validation and testing	*[Insert original ideas]*

[Insert original brainstorming board here]

Figure 2.2: Brainstorming board (original board to be inserted)

2.4 Idea Prioritization

The nine prioritized ideas and their priority decisions come from the original Ideation Phase documentation and are to be inserted in Table 2.4. The rows are numbered only to reserve the space; no idea or decision has been assumed.

Table 2.4: Prioritized ideas and priority decisions

No.	Idea	Priority decision
1	*[Insert original idea]*	*[Insert original decision]*
2	*[Insert original idea]*	*[Insert original decision]*
3	*[Insert original idea]*	*[Insert original decision]*
4	*[Insert original idea]*	*[Insert original decision]*
5	*[Insert original idea]*	*[Insert original decision]*
6	*[Insert original idea]*	*[Insert original decision]*
7	*[Insert original idea]*	*[Insert original decision]*
8	*[Insert original idea]*	*[Insert original decision]*
9	*[Insert original idea]*	*[Insert original decision]*

Why the selected ideas suit Incident Lifecycle Automation. Every brainstorming group maps to a stage of the incident lifecycle and to a functional requirement, as shown in Table 2.5. Ideas that are prioritized therefore contribute directly to the scope defined in Section 1.5 rather than to features outside the incident process.

Table 2.5: Mapping of idea groups to functional requirements

Idea group	Related functional requirements
Incident Setup	FR-1 Service & Service Offering; FR-2 Incident Record Creation
Classification & Routing	FR-3 Incident Classification; FR-5 Reassignment & Escalation
Knowledge Support	FR-4 Knowledge Management & Agent Assist
Related Records	FR-7 Emergency Change; FR-8 Child Incident; FR-11 SLA & Related Records
Resolution & Knowledge	FR-9 Resolution & Closure; FR-10 Knowledge Article Creation
Validation	FR-12 Final Validation & Testing

3. Requirement Analysis

3.1 Customer Journey Map

The Customer Journey Map follows the Service Desk Agent, supported by the Level 2 Technician, through five stages: Entice, Enter, Engage, Exit, and Reuse. Table 3.1 explains each stage in terms of the incident lifecycle. The stage names are those of the original journey map; the explanations are drawn from the documented lifecycle and should be adjusted if the original map differs.

Table 3.1: Customer journey stages

Stage	Service Desk Agent	Level 2 Technician (supporting)	Lifecycle link
Entice	An issue is reported and the agent recognises that an incident record is required.	Not yet involved.	Service and Service Offering setup (FR-1)
Enter	Opens the Incident Management interface, creates the incident, and enters the required information.	Not yet involved.	FR-2 Incident Record Creation
Engage	Classifies the incident, uses Knowledge and Agent Assist, assigns or escalates, and links change or child incidents.	Reviews the escalated incident and its context, investigates, and records findings.	FR-3 to FR-8
Exit	Records Cause, Resolution, Resolution Details, and closure information, and closes the incident.	Returns the incident for resolution and closure.	FR-9 Resolution & Closure
Reuse	Creates a Knowledge Article from useful troubleshooting information and checks related records and SLA.	Findings become available for future incidents through the article.	FR-10, FR-11, FR-12

[Insert original Customer Journey Map here]

Figure 3.1: Customer Journey Map (original map to be inserted)

3.2 Functional Requirements

Twelve functional requirements describe what the solution must do. Each one corresponds to a step of the functional flow in Section 4.4 and is tested in Chapter 7.

Table 3.2: Functional requirements

ID	Requirement	Description	Expected outcome
FR-1	Service & Service Offering	Set up the Service and Service Offering records and associate them with the incident.	Service context is available for classification, assignment, and SLA visibility.
FR-2	Incident Record Creation	Create a new incident in the Incident Management interface and enter the required incident information.	The incident contains enough information for triage and routing.
FR-3	Incident Classification	Classify the incident using Category, Subcategory, Urgency, Service, and Configuration Item.	Classification is completed consistently.
FR-4	Knowledge Management & Agent Assist	Make Knowledge and Agent Assist information available during troubleshooting.	Agents can use existing knowledge to support investigation.
FR-5	Reassignment & Escalation	Assign or reassign the incident and escalate it to Level 2 support when additional support is required.	The incident reaches the appropriate support level.

ID	Requirement	Description	Expected outcome
FR-6	Level 2 Investigation	The Level 2 Technician reviews the escalated incident and its context, investigates the issue, and updates findings.	Escalated incidents retain context for Level 2.
FR-7	Emergency Change	Link an Emergency Change Request to the incident when required.	The emergency change is traceable to the incident.
FR-8	Child Incident	Create or link Child Incidents to represent recurring or related issues under a parent incident.	Related issues are represented and traceable.
FR-9	Resolution & Closure	Document Cause, Resolution, Resolution Details, and closure information before closing the incident.	Resolution and closure information is complete.
FR-10	Knowledge Article Creation	Convert useful troubleshooting information into a Knowledge Article when applicable.	Reusable knowledge is created.
FR-11	SLA & Related Records	Make the SLA and related records (change, child incidents, knowledge) visible from the incident.	SLA and related records are visible and traceable.
FR-12	Final Validation & Testing	Validate the end-to-end workflow through testing and deployment validation.	The workflow passes testing and deployment validation.

3.3 Non-Functional Requirements

Six non-functional requirements describe the qualities expected of the solution. Where the source documents give no figures or configuration, this is stated instead of assumed.

Table 3.3: Non-functional requirements

Requirement	Description
Usability	Service Desk Agents and Level 2 Technicians should be able to follow the lifecycle from the Incident Management interface using consistent field names and the guidance in Section 4.5.
Security	Access to incident, change, knowledge, and related records should be controlled through ServiceNow roles and access controls so that each role sees and updates only what it needs. Specific role and access-control configuration is not documented in the source material.
Reliability	Relationships between incident, change, child incident, knowledge, and SLA records must stay accurate and traceable throughout the lifecycle. This is checked by the end-to-end test TC-12.
Performance	Standard ServiceNow incident operations should respond acceptably for normal service-desk use. No performance targets or measurements are defined in the source documents (see Section 7.6).
Availability	The solution depends on the availability of the ServiceNow instance on which it is configured. No separate hosting or infrastructure is used.
Scalability	Because the solution uses standard Incident, Service, Knowledge, and Change functionality, it can be extended to more services, offerings, and support groups by configuration. Behaviour at scale has not been tested.

3.4 Data Flow Diagram

Incident information moves through the ServiceNow platform from the Service Desk Agent to closure and reuse, as shown in Figure 3.2. The diagram is redrawn from the Functional Flow in the original documentation, which lists the same eleven steps in the same order.

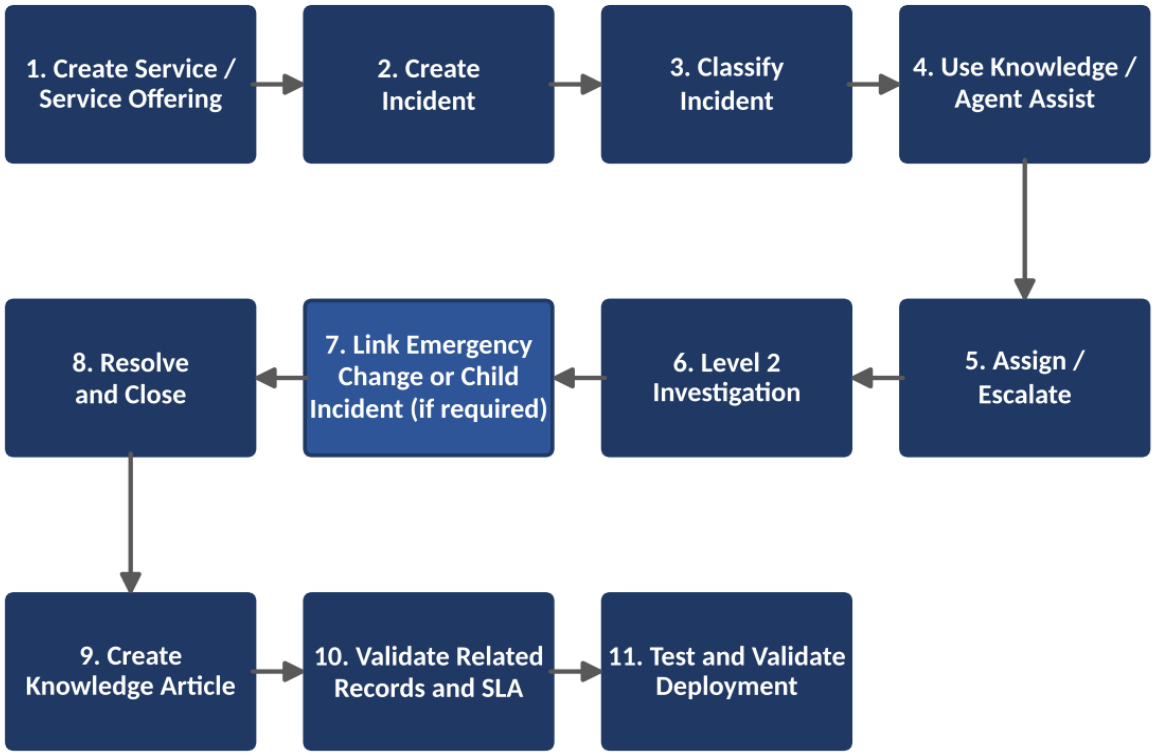


Figure 3.2: Incident lifecycle and information flow (redrawn from the original Functional Flow)

If the original Data Flow Diagram is available, insert it here in addition to or in place of Figure 3.2. It was not part of the supplied files.

Table 3.4 describes the main movements of information between the Service Desk Agent, the ServiceNow components, and Level 2 support.

Table 3.4: Information flows in the incident lifecycle

From	To	Information passed
Service Desk Agent	Incident Management	Required incident information entered when the incident is created.
Service / Service Offering / CI	Incident record	Service, Service Offering, and CI selected for the incident.
Incident record	Knowledge / Agent Assist	Incident context used to find relevant knowledge; knowledge suggestions and articles are returned to the agent.
Incident record	Level 2 Technician	Escalated incident with its existing context. Level 2 returns technical findings and resolution information.
Incident record	Emergency Change / Child Incident	Link between the incident and the related change request or child incident, when required.
Incident record	Resolution and closure fields	Cause, Resolution, Resolution Details, and closure information.
Resolved incident	Knowledge Article	Useful troubleshooting information converted into a reusable article.
Incident and related records	Final validation	Incident, SLA, change, child incident, knowledge, and related records checked against the validation checklist.

3.5 Technology Stack

The solution is centred on the ServiceNow platform. All lifecycle records are held in ServiceNow, and no technology outside the platform is documented in the source material.

Table 3.5: ServiceNow-centred technology stack

Layer	Component	Role in the solution
Platform	ServiceNow instance *[record instance type and release]*	Hosts all incident, service, knowledge, change, and SLA records and the user roles.
User interface	ServiceNow UI / Incident Workspace	Interface used by agents and technicians to work with incidents and related records.
Incident process	Incident Management	Core process and record for creation, classification, investigation, resolution, and closure.
Service context	Service, Service Offering, CMDB / Configuration Item	Identify the affected business service, its offering, and the configuration item.
Knowledge	Knowledge Management and Agent Assist	Provide knowledge during troubleshooting and store Knowledge Articles created from resolved incidents.
Escalation	Assignment and Level 2 support	Route incidents to the appropriate technician or support group.
Change	Change Management (Emergency Change Request)	Handle emergency changes that are linked to incidents.
Relationships	Parent / Child Incident, related records	Represent recurring or related issues and show linked records.
Service levels	SLA	Provide SLA information on the incident.

4. Project Design

4.1 Problem-Solution Fit

The Problem-Solution Fit canvas of the original Project Design Phase has twelve sections. Their titles and content are to be inserted in Table 4.1 from the original document, because they were not part of the supplied files. The way the proposed solution answers the problems of the two primary users is explained after the table.

Table 4.1: Problem-Solution Fit canvas sections

No.	Canvas section	Original content
1	*[Insert section title]*	*[Insert original content]*
2	*[Insert section title]*	*[Insert original content]*
3	*[Insert section title]*	*[Insert original content]*
4	*[Insert section title]*	*[Insert original content]*
5	*[Insert section title]*	*[Insert original content]*
6	*[Insert section title]*	*[Insert original content]*
7	*[Insert section title]*	*[Insert original content]*
8	*[Insert section title]*	*[Insert original content]*
9	*[Insert section title]*	*[Insert original content]*
10	*[Insert section title]*	*[Insert original content]*
11	*[Insert section title]*	*[Insert original content]*
12	*[Insert section title]*	*[Insert original content]*

[Insert original Problem-Solution Fit canvas here]

Figure 4.1: Problem-Solution Fit canvas (original canvas to be inserted)

How the solution addresses the user problems. Table 4.2 relates the needs of the Service Desk Agent (PS-1) and the Level 2 Technician (PS-2) to the parts of the designed solution.

Table 4.2: Problems addressed by the proposed solution

User	Problem faced (from lifecycle requirements)	Solution element	Requirement
Service Desk Agent (PS-1)	Incidents may lack the information needed for triage and routing.	Structured incident creation with required information.	FR-2
Service Desk Agent (PS-1)	Classification may be inconsistent.	Classification by Category, Subcategory, Urgency, Service, and CI.	FR-1, FR-3
Service Desk Agent (PS-1)	Troubleshooting depends on individual knowledge.	Knowledge Management and Agent Assist support.	FR-4
Service Desk Agent (PS-1)	Resolution information may be incomplete and not reused.	Documented Cause, Resolution, Resolution Details, closure information, and Knowledge Article creation.	FR-9, FR-10
Level 2 Technician (PS-2)	Escalated incidents may arrive without context.	Escalation that retains the incident context; Level 2 investigation on the same record.	FR-5, FR-6
Level 2 Technician (PS-2)	Related changes and issues are hard to trace.	Emergency Change and Child Incident links; SLA and related-record visibility.	FR-7, FR-8, FR-11

4.2 Proposed Solution

The proposed solution is described by six parameters. Where the original Project Design Phase wording is not available, the entry states the documented design characteristic and reserves a field for the original text.

Table 4.3: Proposed solution parameters

Parameter	Description
Problem Statement	Inconsistent incident handling affects the Service Desk Agent (PS-1) and the Level 2 Technician (PS-2), as described in Section 2.1. *[Insert original wording if different]*
Idea / Solution Description	A structured Incident Management lifecycle in ServiceNow: create Service and Service Offering, create and classify the incident, use Knowledge and Agent Assist, assign or escalate to Level 2, link Emergency Change or Child Incident when required, resolve and close, create a Knowledge Article, validate related records and SLA, and test and validate deployment.
Novelty / Uniqueness	The design connects the whole lifecycle in one flow, from service setup to knowledge reuse, with traceable links between incident, change, child incident, knowledge, and SLA records. *[Insert original novelty statement]*
Social Impact / Customer Satisfaction	A consistent process and reusable knowledge are expected to give users of IT services more predictable support. This is an expected benefit and has not been measured. *[Insert original statement]*
Business Model	*[Not documented in the supplied material – insert original statement]*
Scalability	The solution relies on standard ServiceNow Incident, Service, Knowledge, and Change functionality and can be extended to further services and support groups by configuration. Scalability has not been tested.

4.3 Solution Architecture

Figure 4.2 shows the architecture of the solution. The Service Desk Agent works through the ServiceNow UI / Incident Workspace. Incident Management is the central component; the service, knowledge, change, and parent/child components supply context to the incident, and the lifecycle ends with resolution, knowledge creation, related-record checks, and final validation. The diagram is redrawn from the component list and functional flow of the original documentation.

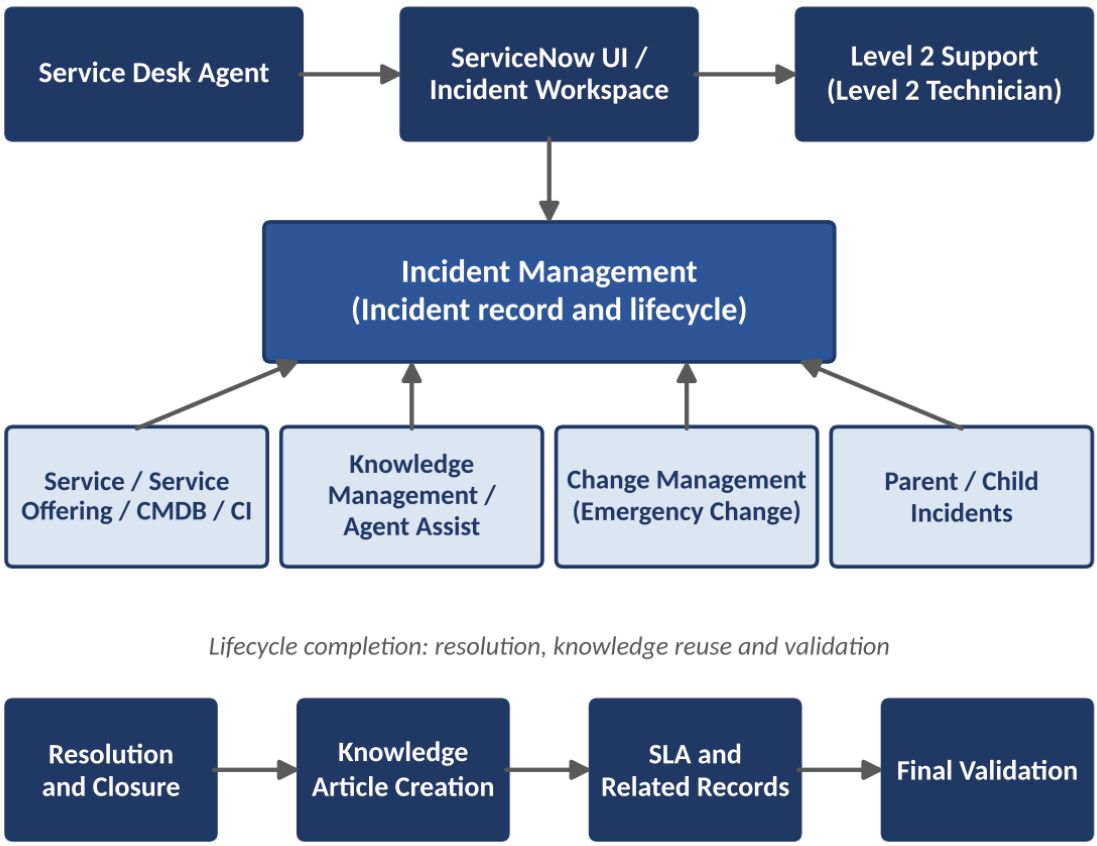


Figure 4.2: Solution architecture of Incident Lifecycle Automation (redrawn from the original description)

If an original architecture diagram exists, insert it in place of or beside Figure 4.2. It was not part of the supplied files.

Table 4.4: Roles of the architecture components

Component	Role in the architecture
Service Desk Agent	Primary user who creates, classifies, assigns, investigates, and updates incidents.
ServiceNow UI / Incident Workspace	Interface through which agents and technicians work with incidents, knowledge, and related records.
Incident Management	Central process and record that holds the incident and its lifecycle information.
Service / Service Offering / CMDB / CI	Provide the service and configuration-item context selected on the incident.
Knowledge Management / Agent Assist	Provide knowledge during troubleshooting and store reusable articles.
Level 2 Support	Handles escalated incidents that need further technical investigation.
Change Management	Holds Emergency Change Requests that are linked to incidents when required.
Parent / Child Incidents	Represent recurring or related issues and their relationship to a parent incident.
SLA and Related Records	Show SLA information and linked records for prioritisation and verification.
Resolution and Closure	Hold Cause, Resolution, Resolution Details, and closure information.
Knowledge Article Creation	Converts useful troubleshooting information into a reusable Knowledge Article.
Final Validation	Confirms that the incident, SLA, change, child incident, knowledge, and related records are correct and the workflow passes testing.

4.4 Architecture and Data Flow Explanation

The components interact in sequence during the incident lifecycle. Table 4.5 follows the eleven steps of the functional flow shown in Figure 3.2 and names the components involved at each step.

Table 4.5: Component interaction through the incident lifecycle

Step	Lifecycle stage	Components involved	Interaction and outcome
1	Create Service / Service Offering	Service, Service Offering	The service context that incidents will refer to is set up and associated.
2	Create Incident	Service Desk Agent, UI / Incident Workspace, Incident Management	The agent creates the incident and enters the required information.
3	Classify Incident	Incident Management, Service, CI	Category, Subcategory, Urgency, Service, and CI are selected.
4	Use Knowledge / Agent Assist	Knowledge Management, Agent Assist	The agent uses available knowledge during troubleshooting.
5	Assign / Escalate	Incident Management, Level 2 Support	The incident is assigned or escalated when additional support is required.
6	Level 2 Investigation	Level 2 Technician, Knowledge, related records	The technician reviews the context, investigates, and records findings.
7	Link Emergency Change or Child Incident	Change Management, Parent / Child Incidents	Related change or child-incident records are linked when required.
8	Resolve and Close	Incident Management	Cause, Resolution, Resolution Details, and closure information are documented.
9	Create Knowledge Article	Knowledge Management	Useful troubleshooting information becomes a Knowledge Article.
10	Validate Related Records and SLA	SLA, related records	Incident, SLA, change, child incident, and knowledge records are verified.
11	Test and Validate Deployment	Project Team	The end-to-end workflow is tested and deployment readiness is validated.

Incident classification fields

Table 4.6: Incident classification fields

Field	Purpose
Category	Identifies the general type of the issue.
Subcategory	Gives a more specific type within the selected category.
Urgency	Shows how quickly the issue needs attention.
Service	Identifies the business service affected.
Configuration Item (CI)	Identifies the specific item in the configuration database that is affected.

Lifecycle requirements

The Functional Design Document sets the following lifecycle requirements:

- The incident should contain sufficient information for triage and routing.
- Classification should be completed consistently.
- Knowledge and Agent Assist should support troubleshooting.
- Escalated incidents should retain context for Level 2.

- Emergency changes should be traceable to the incident when required.
- Recurring or related issues should be represented using child incidents.
- Cause, Resolution, Resolution Details, and closure information should be documented.
- Useful troubleshooting information should be eligible for conversion into a Knowledge Article.

4.5 User Guide

The User Guide of the original documentation explains how each role carries out its part of the lifecycle. The steps are retained below.

Service Desk Agent

1. Open the Incident Management interface.
2. Create a new incident and enter the required incident information.
3. Select Category, Subcategory, Urgency, Service, and CI.
4. Use available Knowledge or Agent Assist information during troubleshooting.
5. Assign or escalate the incident when additional support is required.
6. Link an Emergency Change or create/link a Child Incident when appropriate.
7. Record Cause, Resolution, Resolution Details, and closure information.
8. Validate the incident and related records before closure.

Level 2 Technician

1. Review the escalated incident and its existing context.
2. Investigate the technical issue.
3. Review related knowledge and associated records.
4. Update findings and resolution information.
5. Coordinate related change or child-incident activity when required.
6. Return the incident for resolution/closure after investigation.

Change Stakeholders

1. Review emergency change requests linked to incidents.
2. Validate change context and relationship to the incident.
3. Support change completion and traceability.

Important note (from the original User Guide): the exact navigation labels, field values, and UI behaviour depend on the final ServiceNow configuration. This guide should be updated with screenshots after the implementation is executed.

5. Project Planning and Scheduling

5.1 Methodology and Planning Objectives

The project is planned with the Agile/Scrum methodology in two sprints. Work is listed in a product backlog, written as user stories, estimated in story points, and assigned to sprints. Progress is followed through a project tracker and a burndown chart.

Note: the Project Planning Phase document was not among the supplied files. Sprint dates, story-point values, velocity, and completion status have therefore **not** been entered, and no values have been assumed. The tables below keep the original structure and mark the fields to be completed with the original figures and calculations.

The planning objectives are to:

- organise the twelve functional requirements into a product backlog of user stories;
- estimate the effort of each story using story points on a Fibonacci scale;
- distribute the stories over two sprints with a clear objective for each sprint; and
- monitor progress using the project tracker and the burndown chart.

5.2 Product Backlog and User Stories

Table 5.1 lists one user story for each functional requirement. The story wording is drafted from the requirements and the User Guide; replace it with the original stories where they differ. Story points and sprint allocation are left for the original values.

Table 5.1: Product backlog and user stories

ID	As a	I want to	FR	Story points	Sprint
USN-1	Service Desk Agent	set up a Service and Service Offering and associate them with incidents, so that service context is available.	FR-1	*[supply]*	*[supply]*
USN-2	Service Desk Agent	create an incident with the required information, so that it can be triaged and routed.	FR-2	*[supply]*	*[supply]*
USN-3	Service Desk Agent	classify the incident by Category, Subcategory, Urgency, Service, and CI, so that it is handled consistently.	FR-3	*[supply]*	*[supply]*
USN-4	Service Desk Agent	use Knowledge and Agent Assist during troubleshooting, so that known solutions are reused.	FR-4	*[supply]*	*[supply]*
USN-5	Service Desk Agent	assign or escalate the incident to Level 2 when more support is required.	FR-5	*[supply]*	*[supply]*
USN-6	Level 2 Technician	review the escalated incident with its context and record my findings.	FR-6	*[supply]*	*[supply]*
USN-7	Service Desk Agent	link an Emergency Change to the incident, so that the change can be reviewed by Change Stakeholders.	FR-7	*[supply]*	*[supply]*
USN-8	Service Desk Agent	create or link a Child Incident for recurring or related issues.	FR-8	*[supply]*	*[supply]*
USN-9	Service Desk Agent	record Cause, Resolution, Resolution Details, and closure information before closing.	FR-9	*[supply]*	*[supply]*
USN-10	Service Desk Agent	create a Knowledge Article from useful troubleshooting information.	FR-10	*[supply]*	*[supply]*
USN-11	Service Desk Agent	view SLA and related records from the incident.	FR-11	*[supply]*	*[supply]*
USN-12	Project Team	validate the end-to-end workflow through testing and deployment validation.	FR-12	*[supply]*	*[supply]*

5.3 Story-Point Estimation and Fibonacci Scale

Story points express the relative size of a story, combining effort, complexity, and uncertainty. A Fibonacci-style scale (1, 2, 3, 5, 8, 13) is commonly used because the growing gaps between values reflect the growing uncertainty of larger stories. The meaning of each value is given in Table 5.2.

Table 5.2: Fibonacci story-point scale

Story points	Relative size
1	Very small and well understood
2	Small
3	Small to medium
5	Medium
8	Large, with some uncertainty
13	Very large; consider splitting the story

5.4 Sprint Schedule and Objectives

The original plan uses two sprints. Their objectives, dates, and totals are to be inserted from the original Project Planning Phase document.

Table 5.3: Sprint schedule and objectives

Sprint	Stories included	Sprint objective	Start date	End date	Total story points
Sprint-1	*[supply]*	*[supply]*	*[supply]*	*[supply]*	*[supply]*
Sprint-2	*[supply]*	*[supply]*	*[supply]*	*[supply]*	*[supply]*

5.5 Planned Velocity and Project Tracker

Velocity is the number of story points a team completes in a sprint. For a sprint, the average velocity per day is calculated as: **Average velocity = Total story points ÷ Sprint duration (days)**. The original calculations are to be inserted in Table 5.4; no value has been assumed.

Table 5.4: Planned velocity and project tracker

Sprint	Total story points	Duration (days)	Planned velocity	Story points completed	Sprint release date
Sprint-1	*[supply]*	*[supply]*	*[supply]*	*[supply]*	*[supply]*
Sprint-2	*[supply]*	*[supply]*	*[supply]*	*[supply]*	*[supply]*

5.6 Planned Burndown Chart

The burndown chart plots the story points remaining against the days of the sprint. The planned line shows how the work is expected to reduce; the actual line is added as work is completed. The original chart is to be inserted.

[Insert original planned burndown chart here]

Figure 5.1: Planned burndown chart (original chart to be inserted)

6. Implementation / Project Development

6.1 Implementation Overview and Status

This chapter describes the development of the solution in the order of the incident lifecycle. For every implementation area it gives the objective, the implementation approach, the expected system behaviour, the implementation status, and the supporting evidence.

The source documentation describes the designed/planned solution and states that no actual execution evidence was provided. The **approach** given for each area is therefore the design-level approach taken from the documentation. Exact field values, navigation, and configuration settings are marked as *[to be supplied]* and must be added from the real ServiceNow instance. Every status below is **Planned – Not Verified**; a status is changed to Implemented or Partially Implemented only when a screenshot or test record is attached.

Table 6.1: Summary of implementation areas

Section	Implementation area	Requirement	Status	Evidence
6.2	Service and Service Offering configuration	FR-1	Planned – Not Verified	Figure 8.2
6.3	Incident record creation	FR-2	Planned – Not Verified	Figures 8.1, 8.3
6.4	Incident classification	FR-3	Planned – Not Verified	Figure 8.4
6.5	Knowledge Management / Agent Assist	FR-4	Planned – Not Verified	Figure 8.5
6.6	Assignment and escalation	FR-5	Planned – Not Verified	Figure 8.6
6.7	Level 2 investigation	FR-6	Planned – Not Verified	Figure 8.6
6.8	Emergency Change linkage	FR-7	Planned – Not Verified	Figure 8.7
6.9	Child Incident relationships	FR-8	Planned – Not Verified	Figure 8.8
6.10	Resolution and closure	FR-9	Planned – Not Verified	Figure 8.9
6.11	Knowledge Article creation	FR-10	Planned – Not Verified	Figure 8.10
6.12	SLA and related-record validation	FR-11, FR-12	Planned – Not Verified	Figures 8.11, 8.12

6.2 Service and Service Offering Configuration

Objective	Set up the Service and Service Offering that incidents will refer to, and associate them with incidents (FR-1).
Implementation approach	Create the Service and its Service Offering records and make them selectable on the incident. Exact records, names, and values: *[to be supplied]*.
Expected system behaviour	The correct Service and Service Offering can be selected on an incident and provide service context for classification and SLA visibility.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.2]*

6.3 Incident Record Creation

Objective	Create an incident that contains sufficient information for triage and routing (FR-2).
Implementation approach	The Service Desk Agent opens the Incident Management interface, creates a new incident, and enters the required incident information. Mandatory fields: *[to be supplied]*.
Expected system behaviour	A new incident record is created with its required information and can proceed to classification.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figures 8.1 and 8.3]*

6.4 Incident Classification

Objective	Classify each incident consistently (FR-3).
Implementation approach	Select Category, Subcategory, Urgency, Service, and Configuration Item on the incident. Choice values and CI data used: *[to be supplied]*.
Expected system behaviour	The incident holds consistent classification values that support routing and later analysis.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.4]*

6.5 Knowledge Management / Agent Assist

Objective	Support troubleshooting with existing knowledge (FR-4).
Implementation approach	Use Knowledge Management and Agent Assist from the incident during troubleshooting. Knowledge base and articles used: *[to be supplied]*.
Expected system behaviour	Knowledge or Agent Assist information is available to the agent as designed while the incident is investigated.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.5]*

6.6 Assignment and Escalation

Objective	Route the incident to the right support level and escalate to Level 2 when needed (FR-5).
Implementation approach	The agent assigns or escalates the incident when additional support is required. Level 2 group and members: *[to be supplied]*.
Expected system behaviour	The incident is assigned to Level 2 support and keeps its existing context.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.6]*

6.7 Level 2 Investigation

Objective	Let the Level 2 Technician investigate the escalated incident and record findings (FR-6).
Implementation approach	The technician reviews the incident and context, investigates the technical issue, reviews related knowledge and records, updates findings, and returns the incident for resolution/closure.
Expected system behaviour	Technical findings and resolution information are recorded on the same incident.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.6]*

6.8 Emergency Change Linkage

Objective	Make an emergency change traceable to the incident when required (FR-7).
Implementation approach	Link an Emergency Change Request to the incident. Change Stakeholders review the linked change. Change record details: *[to be supplied]*.
Expected system behaviour	The Emergency Change appears as a related record of the incident and can be traced in both directions.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.7]*

6.9 Child Incident Relationships

Objective	Represent recurring or related issues using child incidents (FR-8).
Implementation approach	Create or link a Child Incident to a parent incident where appropriate. Parent and child incident numbers: *[to be supplied]*.
Expected system behaviour	The parent–child relationship is visible from both incidents.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.8]*

6.10 Resolution and Closure

Objective	Document the resolution completely before the incident is closed (FR-9).
Implementation approach	Record Cause, Resolution, Resolution Details, and closure information, and validate the incident and related records before closure.
Expected system behaviour	The incident cannot be treated as complete without its resolution and closure information, which remain visible on the record.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.9]*

6.11 Knowledge Article Creation

Objective	Turn useful troubleshooting information into reusable knowledge (FR-10).
Implementation approach	Create a Knowledge Article from the resolved incident when it is applicable. Article title and knowledge base: *[to be supplied]*.
Expected system behaviour	A Knowledge Article exists and is available for future incidents.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figure 8.10]*

6.12 SLA and Related-Record Validation

Objective	Verify SLA and related-record visibility and validate the end-to-end workflow (FR-11, FR-12).
Implementation approach	Check the incident, SLA, change, child incident, knowledge, and related records against the validation checklist in Section 7.7. SLA definitions used: *[to be supplied]*.
Expected system behaviour	SLA information and all related records are visible and traceable from the incident, and the end-to-end workflow passes.
Actual implementation status	Planned – Not Verified. The design is documented; no implementation evidence has been supplied.
Supporting evidence	*[Insert actual ServiceNow screenshot – Figures 8.11 and 8.12]*

7. Functional and Performance Testing

7.1 Testing Approach

Testing follows the validation activities defined in the original Testing and Deployment Validation document. Each functional requirement is covered by at least one test case, and the full lifecycle is checked by an end-to-end scenario. The traceability between requirements and test cases is shown in Table 7.1.

Planned and completed tests. All test cases in this chapter are **planned**. The original documentation states that execution results are not asserted because no execution evidence was provided. The Actual Result column is therefore marked “Not recorded”, the Status is “Planned – Not Executed”, and the Evidence column holds a placeholder. No test result, metric, or bug fix has been assumed. When a test is executed, the project team should record the observed result, change the status to Passed or Failed, and attach the screenshot.

Table 7.1: Traceability between functional requirements and test cases

Requirement	Test case	Requirement	Test case
FR-1 Service & Service Offering	TC-01	FR-7 Emergency Change	TC-07
FR-2 Incident Record Creation	TC-02	FR-8 Child Incident	TC-08
FR-3 Incident Classification	TC-03	FR-9 Resolution & Closure	TC-09
FR-4 Knowledge Management & Agent Assist	TC-04	FR-10 Knowledge Article Creation	TC-10
FR-5 Reassignment & Escalation	TC-05	FR-11 SLA & Related Records	TC-11
FR-6 Level 2 Investigation	TC-06	FR-12 Final Validation & Testing	TC-12

7.2 Functional Testing

Table 7.2 lists the functional test cases. Preconditions, steps, and expected results are taken from the User Guide and the Validation Checklist of the original documentation.

Table 7.2: Functional test cases

ID	Test scenario	Preconditions	Test steps	Expected result	Actual result	Status	Evidence
TC-01	Validate Service and Service Offering configuration	Service and Service Offering records are set up.	1. Open the incident. 2. Select the Service and Service Offering.	Service and Service Offering are correct and selectable.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-02	Validate incident creation and mandatory information	Agent has access to the Incident Management interface.	1. Create a new incident. 2. Enter the required information. 3. Save.	Incident is created with the required information.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-03	Validate classification	Incident is created.	1. Select Category, Subcategory, Urgency, Service, and CI. 2. Save.	Incident is classified consistently with all five values.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-04	Validate Knowledge Management / Agent Assist usage	Knowledge content is available.	1. Open the classified incident. 2. Use Knowledge / Agent Assist.	Knowledge / Agent Assist information is available as designed.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-05	Validate assignment and Level 2 escalation	Incident is under investigation; Level 2 support is available.	1. Assign or escalate the incident to Level 2.	Incident is assigned to Level 2 support.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-06	Validate Level 2 investigation and retained context	Incident is escalated.	1. Open the escalated incident as Level 2. 2. Review context. 3. Update findings.	Existing context is retained and findings are recorded.	Not recorded	Planned – Not Executed	*[Insert evidence]*

ID	Test scenario	Preconditions	Test steps	Expected result	Actual result	Status	Evidence
TC-07	Validate Emergency Change relationship	Incident requires an emergency change.	1. Link an Emergency Change Request to the incident.	Emergency Change is linked and traceable.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-08	Validate Child Incident relationship	A related or recurring issue exists.	1. Create or link a Child Incident to the parent.	Parent–child relationship is visible.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-09	Validate resolution and closure fields	Investigation is complete.	1. Enter Cause, Resolution, Resolution Details, and closure information. 2. Close.	Resolution and closure information is complete.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-10	Validate Knowledge Article creation	Incident is resolved with useful information.	1. Create a Knowledge Article from the incident.	Knowledge Article is created.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-11	Validate SLA and related-record visibility	Incident has related records.	1. Open the incident. 2. Check SLA and related records.	SLA and related records are visible and traceable.	Not recorded	Planned – Not Executed	*[Insert evidence]*
TC-12	End-to-end scenario	All configuration is complete.	Create incident → classify → use knowledge → escalate to Level 2 → link change/child incident → document cause and resolution → close → create knowledge → verify records and SLA.	Workflow passes end to end.	Not recorded	Planned – Not Executed	*[Insert evidence]*

7.3 AI / Predictive Intelligence Testing

Not applicable. The source documentation does not describe any AI or Predictive Intelligence feature, so no such tests are defined. Agent Assist is tested only as a knowledge-support feature in TC-04.

7.4 Machine Learning Evaluation

Not applicable. No machine-learning model, dataset, or evaluation metric is described in the source documentation.

7.5 Integration Testing

Integration testing checks that the records of the lifecycle work together. These checks are covered by the functional test cases and are planned, not executed.

Table 7.3: Integration checks

Integration point	Covered by	Status
Incident and Service / Service Offering / CI	TC-01, TC-03	Planned – Not Executed
Incident and Knowledge / Agent Assist	TC-04	Planned – Not Executed
Incident and Level 2 support	TC-05, TC-06	Planned – Not Executed
Incident and Emergency Change	TC-07	Planned – Not Executed
Parent and Child Incident	TC-08	Planned – Not Executed
Incident and Knowledge Article	TC-10	Planned – Not Executed
Incident and SLA / related records	TC-11, TC-12	Planned – Not Executed

7.6 Performance Testing

The source documentation defines no performance targets, load profiles, or measurements, so none are stated here. Table 7.4 reserves space for performance results if the project team decides to measure them.

Table 7.4: Performance measurements

Measure	Target	Observed result	Status
[Define measure, e.g. time to open or save an incident]	*[supply]*	*[supply]*	Not defined

7.7 User Acceptance Testing and Deployment Readiness

User acceptance and deployment readiness are judged against the Deployment Readiness list and the Validation Checklist of the original documentation. No item has been confirmed, because no execution evidence was supplied.

Table 7.5: Deployment readiness and validation checklist

Criterion	Related test	Status
All planned functional tests completed	TC-01 to TC-12	Not Verified
No unresolved critical defects	Defect log (Table 7.6)	Not Verified
Incident, change, child incident, knowledge, SLA, and related-record relationships verified	TC-07, TC-08, TC-10, TC-11	Not Verified
User acceptance evidence captured	Chapter 8	Not Verified
Deployment validation completed	TC-12	Not Verified
Incident record is created and classified	TC-02, TC-03	Not Verified
Service and Service Offering are correct	TC-01	Not Verified
SLA / related records are visible and traceable	TC-11	Not Verified
Emergency Change is linked where applicable	TC-07	Not Verified
Child Incident is linked where applicable	TC-08	Not Verified
Knowledge / Agent Assist information is available as designed	TC-04	Not Verified
Resolution and closure information is complete	TC-09	Not Verified
Knowledge Article is created when applicable	TC-10	Not Verified
End-to-end workflow passes testing and deployment validation	TC-12	Not Verified

7.8 Defect Log

No bug IDs, defects, or fixes appear in the source documentation, so none are recorded. Use Table 7.6 to record defects found during execution.

Table 7.6: Defect log

Bug ID	Related test	Description	Severity	Status
[none recorded]	*[ ]*	*[ ]*	*[ ]*	*[ ]*

8. Results and Output Screenshots

This chapter groups the screenshots that provide the visual evidence for the lifecycle. **No screenshot was supplied with the original documentation, and none has been generated.** Each placeholder must be replaced with a genuine ServiceNow screenshot. Until that is done, a placeholder is not evidence and the related capability stays Planned – Not Verified.

8.1 ServiceNow Incident Interface

[Insert actual ServiceNow screenshot here]

Figure 8.1: ServiceNow Incident Interface

What it should demonstrate: the Incident Management interface (or Incident Workspace) from which the Service Desk Agent works. **Referenced in:** Section 6.3.

8.2 Service and Service Offering

[Insert actual ServiceNow screenshot here]

Figure 8.2: Service and Service Offering

What it should demonstrate: the Service and Service Offering records and their association with the incident. **Referenced in:** Section 6.2, TC-01.

8.3 Incident Creation

[Insert actual ServiceNow screenshot here]

Figure 8.3: Incident Creation

What it should demonstrate: a newly created incident with its required information entered. **Referenced in:** Section 6.3, TC-02.

8.4 Incident Classification

[Insert actual ServiceNow screenshot here]

Figure 8.4: Incident Classification

What it should demonstrate: the incident with Category, Subcategory, Urgency, Service, and CI selected. **Referenced in:** Section 6.4, TC-03.

8.5 Knowledge Assistance

[Insert actual ServiceNow screenshot here]

Figure 8.5: Knowledge Assistance

What it should demonstrate: Knowledge Management or Agent Assist information shown during troubleshooting. **Referenced in:** Section 6.5, TC-04.

8.6 Level 2 Assignment

[Insert actual ServiceNow screenshot here]

Figure 8.6: Level 2 Assignment

What it should demonstrate: the incident assigned or escalated to Level 2 support, with its context retained and the technician's findings. **Referenced in:** Sections 6.6 and 6.7, TC-05, TC-06.

8.7 Emergency Change

[Insert actual ServiceNow screenshot here]

Figure 8.7: Emergency Change

What it should demonstrate: the Emergency Change Request linked to the incident. **Referenced in:** Section 6.8, TC-07.

8.8 Child Incident

[Insert actual ServiceNow screenshot here]

Figure 8.8: Child Incident

What it should demonstrate: the parent incident with its linked Child Incident. **Referenced in:** Section 6.9, TC-08.

8.9 Resolution and Closure

[Insert actual ServiceNow screenshot here]

Figure 8.9: Resolution and Closure

What it should demonstrate: Cause, Resolution, Resolution Details, and closure information on the incident. **Referenced in:** Section 6.10, TC-09.

8.10 Knowledge Article

[Insert actual ServiceNow screenshot here]

Figure 8.10: Knowledge Article

What it should demonstrate: the Knowledge Article created from the resolved incident. **Referenced in:** Section 6.11, TC-10.

8.11 SLA and Related Records

[Insert actual ServiceNow screenshot here]

Figure 8.11: SLA and Related Records

What it should demonstrate: SLA information and related records visible and traceable from the incident. **Referenced in:** Section 6.12, TC-11.

8.12 Testing Results

[Insert actual ServiceNow screenshot here]

Figure 8.12: Testing Results

What it should demonstrate: the executed test cases and results, including the end-to-end scenario. **Referenced in:** Chapter 7, TC-12.

9. Advantages and Limitations

9.1 Advantages

The advantages in Table 9.1 are those of the designed solution. They are expected benefits that must be confirmed by execution evidence.

Table 9.1: Advantages of the proposed solution

Advantage	Explanation
Structured incident lifecycle	One defined path from creation to closure gives every incident the same treatment.
Consistent classification	Category, Subcategory, Urgency, Service, and CI are captured on every incident.
Improved routing and escalation	Clear assignment and Level 2 escalation help incidents reach the right team.
Knowledge-assisted investigation	Knowledge Management and Agent Assist support troubleshooting with existing solutions.
Better Level 2 context	Escalated incidents keep their context, so technicians need not repeat questions.
Related-record traceability	Emergency Changes and Child Incidents are linked and can be traced from the incident.
Complete resolution documentation	Cause, Resolution, Resolution Details, and closure information are recorded.
Knowledge reuse	Useful troubleshooting information becomes a Knowledge Article for future incidents.
SLA visibility	SLA information and related records can be viewed from the incident.

9.2 Limitations

The limitations in Table 9.2 are general constraints of the platform and of the project scope. They are not problems observed during execution.

Table 9.2: Limitations and constraints

Limitation	Type	Description
No execution evidence yet	Project scope	The source documentation states that execution results are not asserted. All capabilities remain Planned – Not Verified until evidence is added.
Dependence on instance configuration	Platform	Navigation labels, field values, and UI behaviour depend on the final ServiceNow configuration.
Dependence on knowledge content	General	The value of Knowledge and Agent Assist depends on the quality and coverage of the available knowledge articles.
SLA behaviour depends on definitions	Platform	SLA information is available only for the SLA definitions that are configured on the instance.
Roles and access not documented	Project scope	Specific user roles, groups, and access controls are not described in the source material.
Performance and scale not evaluated	Project scope	No performance or load testing is defined.
Reliance on agent actions	General	The documented procedures rely on agents to classify, escalate, and link records. Further automation is listed in Chapter 11.
Limited scope	Project scope	No external dataset, integration, or machine-learning component is documented.

10. Conclusion and Learning Outcomes

10.1 Conclusion

The project defines a structured Incident Management lifecycle in ServiceNow. The lifecycle runs from Service and Service Offering setup through incident creation, classification, knowledge-assisted investigation,

assignment and escalation, Level 2 investigation, emergency change and child incident linkage, resolution and closure, Knowledge Article creation, and final validation. Five roles take part: the Service Desk Agent, Level 2 Technician, Service Owner / Support Stakeholders, Change Stakeholders, and Project Team.

The report documents the objectives, scope, twelve functional requirements, six non-functional requirements, architecture, information flow, user procedures, test cases, and configuration reference of the solution. A consistent reference of this kind supports standardized handling, better classification, improved Level 2 context, traceability, and knowledge reuse.

Planned capabilities and verified implementation. The documented capabilities are the designed ones. The source documentation provides no execution evidence, so no capability is reported as verified, and all test cases are planned. When screenshots and test records are added, the status of each area in Chapter 6 and each test case in Chapter 7 can be updated.

10.2 Challenges and Solutions

Only challenges supported by the project documents are listed. No implementation challenges were recorded in the supplied material; the project team may add any that arose during implementation.

Table 10.1: Challenges and solutions

Challenge	Cause	Response in this report
Lack of actual execution evidence	The project documentation materials did not include execution results or screenshots.	Capabilities are labelled Planned – Not Verified, and clearly marked placeholders are provided for evidence.
Missing supporting phase documents	The Ideation, Requirement, Design, and Planning documents were not supplied with the original file.	Original headings and structure are kept and the missing content is marked *[to be supplied]*, not invented.
[Add implementation challenges, if any]	*[ ]*	*[ ]*

10.3 Learning Outcomes

The project provides the following learning outcomes:

- Understanding of ServiceNow Incident Management and the stages of the incident lifecycle.
- Understanding of how incident workflows and records are configured and used.
- Understanding of the relationship between Service, Service Offering, and Configuration Item.
- Understanding of Knowledge Management and Agent Assist in troubleshooting.
- Understanding of assignment and Level 2 escalation.
- Understanding of Emergency Change and Parent / Child Incident relationships.
- Practice in planning tests, validating requirements, and documenting a project with a clear distinction between planned and verified work.
- Technical problem-solving through structuring a lifecycle from user needs to validation.

11. Future Scope

The enhancements in Table 11.1 are proposed for future work. They are **not** part of the documented solution and are not implemented.

Table 11.1: Proposed future enhancements

Proposed enhancement	Description
Advanced incident prioritization	Improve how priority is set from urgency, service, and CI information.
Improved automated classification	Reduce manual classification effort and improve consistency.
Enhanced SLA monitoring	Give clearer warnings and tracking of SLA status.
Advanced reporting dashboards	Show incident volumes, resolution, escalation, and SLA information.
Predictive incident analysis	Identify likely recurring incidents and trends.
Improved knowledge recommendations	Suggest more relevant knowledge during troubleshooting.
Further automation of repetitive activities	Automate repetitive steps such as routing, linking, and notification.

12. Appendix

12.1 Setup / Recreation Manual

The steps below allow an administrator to configure the solution on a ServiceNow instance. The source documentation gives no verified navigation paths, so the application area is named instead of an exact path. Confirm each path on the instance and record the actual values used.

Table 12.1: Setup steps

Step	Application area	What to configure	Expected result
1	Users and groups	Identify or create users and groups for Service Desk Agent, Level 2 Technician, and Change Stakeholders. Roles and groups: *[to be supplied]*.	Users are available for assignment, escalation, and change review.
2	Service portfolio	Create the Service and Service Offering records.	Service context can be selected on incidents.
3	CMDB	Make sure the relevant Configuration Items exist.	A CI can be selected on the incident.
4	Incident	Check that the incident form shows Category, Subcategory, Urgency, Service, CI, and the resolution fields.	Incidents can be created and classified.
5	Incident choices	Confirm the Category and Subcategory values. Values: *[to be supplied]*.	Classification is consistent.
6	Knowledge	Make a knowledge base and articles available and enable Agent Assist.	Knowledge support is available during troubleshooting.
7	Assignment	Set up the Level 2 support group and its members.	Incidents can be escalated to Level 2.
8	Change	Check that an Emergency Change can be created and linked to an incident.	The change appears as a related record.
9	Incident relationships	Check that a Child Incident can be linked to a parent incident.	The relationship is visible.
10	SLA	Check the SLA definitions that apply to incidents. Definitions: *[to be supplied]*.	SLA information is visible on the incident.
11	Knowledge from incident	Check that a Knowledge Article can be created from a resolved incident.	A reusable article can be created.
12	Testing	Run TC-01 to TC-12 and capture the screenshots listed in Section 12.6.	Execution evidence is available.

12.2 Configuration Reference

Table 12.2: Documented field groups

Field group	Fields	Used by	Requirement
Classification	Category, Subcategory, Urgency, Service, Configuration Item	Service Desk Agent	FR-3
Service context	Service, Service Offering	Service Desk Agent; Service Owner	FR-1
Resolution and closure	Cause, Resolution, Resolution Details, closure information	Service Desk Agent; Level 2 Technician	FR-9

Table 12.3: Record relationships

Record	Related to	Purpose
Incident	Service / Service Offering / CI	Provides service and configuration context.
Incident	Emergency Change Request	Makes the emergency change traceable to the incident.
Parent Incident	Child Incident	Represents recurring or related issues.
Incident	Knowledge / Knowledge Article	Supports troubleshooting and records reusable knowledge.
Incident	SLA and related records	Shows service-level information and linked records.

Table 12.4: Instance details to record

Item	Value
ServiceNow release / instance type	*[to be supplied]*
Roles and groups used	*[to be supplied]*
Service and Service Offering names	*[to be supplied]*
Knowledge base used	*[to be supplied]*

12.3 Source Code

Not applicable. The source documentation contains no source code, script, or repository reference, and none has been created for this report.

12.4 Dataset Link

Not applicable. The solution operates on ServiceNow records and does not use an external dataset.

12.5 GitHub and Project Demo Links

The supplied document contains no URL, hyperlink, GitHub link, or demo link, so none has been added. If the original project has such links, paste them here exactly as they appear. Do not shorten or edit them.

GitHub	*[leave blank if none]*
Project Demo Link	*[leave blank if none]*
Other reference links	*[leave blank if none]*

12.6 Screenshot Placement Checklist

Table 12.5: Screenshot placement checklist

No.	Figure	Screenshot required	Inserted
1	Figure 8.1	ServiceNow Incident Interface	☐
2	Figure 8.2	Service and Service Offering	☐
3	Figure 8.3	Incident Creation	☐
4	Figure 8.4	Incident Classification	☐
5	Figure 8.5	Knowledge Assistance	☐
6	Figure 8.6	Level 2 Assignment	☐
7	Figure 8.7	Emergency Change	☐
8	Figure 8.8	Child Incident	☐
9	Figure 8.9	Resolution and Closure	☐
10	Figure 8.10	Knowledge Article	☐
11	Figure 8.11	SLA and Related Records	☐
12	Figure 8.12	Testing Results	☐

12.7 Final Verification Checklist

Table 12.6: Final verification checklist

Check	Result
Project title is Incident Lifecycle Automation in ServiceNow throughout	Checked in this report
ServiceNow terminology is used consistently	Checked in this report
Original roles, scope, workflow, and requirements are retained	Checked in this report
Diagrams: flow and architecture were redrawn from the original text; original diagrams were not supplied	Pending – insert originals if available
Configuration details	Pending – placeholders *[to be supplied]*
Test cases defined; none executed	Planned – Not Executed
Screenshots	Pending – placeholders only
Existing links preserved exactly	No links were present; none added
Implementation status of every capability	Planned – Not Verified
Team ID, report date, program name	Pending – to be entered
Student details on title page	Confirm before submission
Table of contents page numbers	Update after inserting screenshots